

Econ 131
Spring 2026
Danny Yagan

Problem Set 2

DUE: 04-01-2026 5PM PT on Gradescope

Student Name:

Student ID:

- Write or type your answers clearly and in dark ink (physical or electronic ink) so that your responses are legible.
- Tag each of your answers on Gradescope so that it is clear what responses are to which questions.
- **Although you may work in groups**, each student must submit individual sets of solutions. You must note the names of other students that you worked with. Write their names here:

1. True/False/Uncertain with Explanation [25 points]

Determine whether each statement is true, false, or uncertain and explain why. Answers with no explanation will receive no points. [5 points each]

- (a) A dividend income tax cannot raise revenue without reducing investment.

- (b) If the income distribution in the top tax bracket becomes thinner (individuals above the bracket threshold become more concentrated at incomes near the threshold), the government should increase the top marginal tax rate.

- (c) If leisure is a normal good, then an increase in non-labor income creates a negative income effect on labor supply.

- (d) A lump-sum tax in period 1 that is independent of savings changes the intertemporal relative price between current and future consumption.

(e) The Pareto parameter for the entire income distribution is $1/2$.

2. Difference-in-differences [15 points]

Recall the graph from Kleven-Landais-Saez (American Economic Review 2013) that we drew in class of the number of Sports and Entertainment foreign workers in Denmark. Recall the regression that we ran to compute the DD estimate of the impact of the 1991 tax reform on the number of top-earning Sports and Entertainment foreign workers in Denmark on average 1991-2005. Suppose that a policymaker asks for the impact of the 1991 tax reform on the number of top-earning Sports and Entertainment foreign workers in Denmark in 2005.

- (a) [5 points] Is the 2005 impact bigger or smaller than the average 1991-2005 impact?

- (b) [10 points] How would you compute the 2005 impact in a single regression? A complete answer should include the regression equation that you would estimate and the data sample you would use. The regression equation should be written in math like in the Lecture 8B slides, not in Stata or other computer code. (Note: There is more than one way to do this regression, but there is a single simple way based on what we have already done in class.)

3. Business Taxation and Investment [26 points]

Consider a firm that chooses capital K to maximize profit. Let $F(K)$ denote the firm's gross profit from using capital, where $F'(K) > 0$ and $F''(K) < 0$. Let r denote the per-unit cost of capital.

Throughout, assume the firm chooses K optimally and interior solutions exist.

- (a) [4 points] Suppose there is **no tax**. The firm solves

$$\max_K F(K) - rK.$$

Write the first-order condition for the firm's optimal choice of capital.

- (b) [5 points] Now suppose the government imposes a **business income tax** at rate τ on **gross profit**, so the firm solves

$$\max_K (1 - \tau)F(K) - rK.$$

Derive the first-order condition. Does this tax increase, decrease, or leave unchanged the firm's desired level of investment? Briefly explain.

- (c) [5 points] Instead suppose the business income tax applies to profits **after deducting capital costs**, so the firm solves

$$\max_K (1 - \tau)[F(K) - rK].$$

Derive the first-order condition. Does this tax affect the firm's desired investment? Briefly explain.

- (d) [4 points] In 2–3 sentences, explain the economic intuition for why the answers in parts (b) and (c) are different.

- (e) [4 points] Historically, many capital costs have not been deductible immediately from taxable profits. Explain why this makes the tax system less like the case in part (c) and more like the case in part (b).

- (f) [4 points] Suppose the tax system is the one from part (b). Now suppose that the government offers an investment tax credit c : for every dollar spent on capital, the firm gets to reduce its tax bill by $\$c$. What value does c need to be in order for the new tax system as a whole to not distort the firm's choice of capital relative to (a)?

4. Labor Income Taxes and Labor Supply [18 points]

An individual chooses labor supply $n \geq 0$. Consumption is

$$c = (1 - \tau)wn + R,$$

where w is the pre-tax wage, τ is the labor income tax rate, and R is non-labor income.

Consider the two utility functions

$$\begin{aligned} \text{(I)} \quad u_1(c, n) &= c - \frac{n^{1+1/\varepsilon}}{1 + 1/\varepsilon}, \\ \text{(II)} \quad u_2(c, n) &= \log(c) - \alpha \frac{n^{1+1/\varepsilon}}{1 + 1/\varepsilon}, \end{aligned}$$

where $\varepsilon > 0$ and $\alpha > 0$.

- (a) [3 points] For case (I), derive the first-order condition for optimal labor supply. Show that labor supply depends on the net-of-tax wage $(1 - \tau)w$.

- (b) [3 points] In case (I), if the tax rate τ rises, is there an income effect on labor supply? Is there a substitution effect? What happens to labor supply overall?

- (c) [3 points] In case (II), if the tax rate τ rises, explain why labor supply is affected by both an income effect and a substitution effect. Do these two effects go in the same direction or opposite directions?

- (d) [3 points] Which case would generate a larger gap between the compensated and uncompensated labor-supply elasticities? Explain briefly.

- (e) [3 points] Suppose the government increases τ but also adjusts R so that the individual can still reach the same utility level as before. Is this change closer to an uncompensated or a compensated tax change? In this situation, which effect on labor supply is being isolated?

- (f) [3 points] In empirical public finance, labor-supply responses are often summarized using the elasticity of labor income with respect to the net-of-tax rate, $1 - \tau$. Based on your answers above, would you expect this elasticity to be larger in absolute value under case (I) or case (II)? Explain briefly.